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PhiliFinds
Software Project Management Plan
Version 01.00

Revision History

Date	Version	Description	Author
02/27/2026	01.00	SPMP Creation	Atayde, Enriquez, Mariquit, Ygot, Davide

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Software Project Management Plan

1 INTRODUCTION

1.1 PURPOSE OF PROJECT MANAGEMENT PLAN

The purpose of this Project Management Plan (PMP) is to define the processes, roles, responsibilities, and procedures required to successfully manage the **PhiliFinds** project, an AI-powered travel planning Progressive Web Application (PWA) for the Philippines.
This plan provides guidance for project execution, monitoring, and control across all phases of the project lifecycle, ensuring that deliverables meet the requirements outlined in the Software Requirements Specification (SRS).

The intended audience of the **PhiliFinds PMP** includes all project stakeholders, such as:

- **Project Sponsor** – Oversees project funding and alignment with business objectives.
- **Senior Leadership** – Reviews project progress, quality, and compliance.
- **Project Team** – Includes developers, UI/UX designers, QA testers, and AI engineers responsible for implementing the system.

The PMP ensures that the project meets its objectives of delivering a secure, responsive, and user-friendly travel itinerary platform while maintaining high quality, performance, and adherence to Philippine data privacy regulations.

2 EXECUTIVE SUMMARY OF PROJECT MANAGEMENT PLAN

The **PhiliFinds Project Management Plan (PMP)** provides a structured approach to delivering an AI-powered travel planning Progressive Web Application (PWA) tailored for travelers in the Philippines.

The project aims to simplify travel itinerary creation by integrating AI-driven recommendations, cost estimation, and schedule optimization into a single, user-friendly platform. Core functionalities include:

- **Account Creation & User Authentication**: Secure onboarding via Supabase.
- **Dynamic Itinerary Builder**: Collects user travel preferences and generates AI-optimized itineraries.
- **Cost Estimator**: Provides real-time trip expense calculations.
- **Itinerary Sharing & Progress Tracking**: Allows users to share and monitor their travel plans.
- **Navigation Assistance & Notifications**: Offers travel guidance and timely reminders.
- **Emergency Support**: Displays emergency hotline information for quick access.
- The PMP covers scope, schedule, quality, risk, communication, resource, and

compliance management to ensure successful project execution. Agile methodology with 2-week sprints will guide development, testing, and deployment.

- The ultimate goal of **PhiliFinds** is to eliminate planning fatigue for travelers, while maintaining security, usability, and high performance standards as outlined in the SRS.

2.1 ASSUMPTIONS/CONSTRAINTS

Assumptions:

- Users will have an active internet connection to access AI services and Supabase backend.
- Users will use modern web browsers (Chrome, Safari, Edge) on mobile or desktop devices.
- Supabase and Vercel platforms will maintain availability for application hosting and data storage.
- AI itinerary engine will generate optimized itineraries within 20 seconds under normal network conditions.

Constraints:

- PhiliFinds operates primarily as a PWA; offline GPS navigation is not supported.
- Direct booking of flights or accommodations is outside the system's scope.
- Supabase free-tier storage limitations must be observed.
- API keys for AI services and maps must remain secure and cannot be exposed to the client side.
- System performance and responsiveness must meet mobile and desktop usability standards outlined in the SRS.

3 SCOPE MANAGEMENT

The **PhiliFinds project** focuses on developing an AI-powered Progressive Web Application (PWA) that simplifies travel planning for users in the Philippines. The project scope covers design, development, testing, and deployment of a secure, responsive, and user-friendly platform capable of:

- **Dynamic Itinerary Generation**: AI-based personalized travel itineraries based on user inputs (preferences, budget, duration, group size, activity level).
- **Cost Estimation**: Real-time calculation of entrance fees, transportation, meals, and activity-related expenses.
- **Time and Logistics Management**: Optimized schedules between destinations to minimize travel time and balance group requirements.
- **Navigation Assistance**: Provides point-to-point directions and public transport suggestions for users without private vehicles.
- **Itinerary Sharing**: Enables users to share itineraries via unique URLs or QR codes. Shared itineraries are read-only for other users.

- **Progress Checklist**: Allows marking completed activities and tracking pending tasks for organized trips.
- **Notifications and Reminders**: Sends alerts for upcoming itinerary activities.
- **Emergency Support**: Displays relevant hotline numbers and allows quick-dial functionality.

Out of Scope:

- Direct booking of flights or accommodations.
- Offline GPS navigation.
- Integration with third-party booking platforms

3.1 WORK BREAKDOWN STRUCTURE

1. **Project Initiation**
   1.1 Requirement gathering and analysis
   1.2 Project planning and approval
2. **UI/UX Design**
   2.1 Wireframes and mockups
   2.2 Responsive PWA design for desktop and mobile
   2.3 Accessibility and usability compliance
3. **Backend Development**
   3.1 Supabase authentication and database setup
   3.2 Implementation of Row-Level Security (RLS) policies
   3.3 API integration for AI itinerary generation and map services
4. **AI Engine Development**
   4.1 Collect and preprocess travel input data
   4.2 Develop itinerary optimization logic
   4.3 Integrate AI engine with frontend and backend
5. **Frontend Development (Next.js)**
   5.1 Dynamic itinerary builder UI
   5.2 Cost estimator and travel logistics components
   5.3 Itinerary Sharing module (URL/QR code generation and display)
   5.4 Progress checklist, notifications, and emergency support UI
6. **Integration and Testing**
   6.1 Functional testing for all modules
   6.2 Performance testing (≤20s AI response)
   6.3 Security testing (RLS, HTTPS, API key protection)
   6.4 Usability and mobile responsiveness testing
7. **Deployment**
   7.1 Staging environment setup
   7.2 Production deployment on Vercel
   7.3 User acceptance testing

8. **Documentation and Training**
   8.1 User manuals and online guides
   8.2 Developer documentation and API references
9. **Maintenance & Support**
   9.1 Bug fixes and patches
   9.2 Performance monitoring

3.2 DEPLOYMENT PLAN

The deployment of **PhiliFinds** will follow a structured, phased approach to ensure smooth rollout and minimal disruption to end users. The plan includes preparation, testing, production launch, and post-deployment support:

1. Environment Assessment

- Verify that target devices (smartphones, tablets, laptops, desktops) meet browser and connectivity requirements.
- Ensure Supabase backend and Vercel hosting platforms are operational and properly configured.
- Validate that all API services (AI itinerary engine, mapping/geolocation APIs) are accessible and functional.

2. Pre-Deployment Preparation

- Set up staging environment for integration testing of all modules, including:
  - Dynamic Itinerary Builder
  - Cost Estimator
  - Navigation Assistance
  - Itinerary Sharing (QR code / URL functionality)
  - Progress Checklist, Notifications, and Emergency Support
- Configure environment variables for secure API keys and database credentials.
- Conduct data migration (if necessary) from test datasets to production-ready formats.

3. Deployment Execution

- Deploy application to **Vercel** production environment with SSL/TLS enabled for secure connections.
- Connect frontend to Supabase backend for authentication, data storage, and Row-Level Security (RLS).
- Ensure Itinerary Sharing module generates unique URLs/QR codes correctly for sharing functionality.
- Perform smoke tests to validate that all major features work as intended in production.

4. User Acceptance and Rollout

- Conduct limited user acceptance testing (UAT) with sample users representing casual tourists, business travelers, and group travelers.
- Address any feedback or defects before full public release.

- Officially release the PWA via public URLs; users can install on their devices using standard PWA installation prompts.

5. Post-Deployment Support

- Monitor application performance and uptime using logging and analytics tools.
- Respond to critical issues promptly following the issue management plan.
- Maintain regular backups of user data in Supabase and monitor security compliance.
- Provide documentation and tutorials for end users to access and share itineraries safely.

Key Considerations:

- Deployment assumes continuous internet connectivity for end users.
- Offline navigation is not supported; users must have an active network connection.
- The system will be updated in stages, with minimal downtime for production users.

3.3 CHANGE CONTROL MANAGEMENT

All changes to the **PhiliFinds project** will follow a simple, controlled process to ensure that modifications do not negatively affect the system's functionality, schedule, or quality. This applies to all features, including **Dynamic Itinerary Builder, Cost Estimator, Itinerary Sharing, Navigation Assistance, Notifications, and Emergency Support**.

Change Control Process:

1. **Submission of Change Request**
   - Any team member or stakeholder can submit a **Change Request** if a feature needs to be added, improved, or fixed.
   - The request must include:
     - Description of the change
     - Reason for the change
     - Which module is affected (e.g., Itinerary Sharing, AI engine, Notifications)
2. **Impact Assessment**
   - The **Project Manager** evaluates how the change affects:
     - **Scope** – Does it add new functionality or modify existing features?
     - **Schedule** – Will it affect deadlines or project milestones?
     - **Resources** – Does it require extra team work or tools?
3. **Approval**
   - The change is approved or rejected by the project team (Project Manager and teammates).
   - Only approved changes are implemented.
4. **Implementation & Testing**
   - Changes are applied in the development environment.
   - Functional testing is done to make sure other features are not broken.
   - For example, changes to **Itinerary Sharing** must still generate correct URLs/QR codes and remain read-only for shared users.
5. **Deployment & Documentation**

    - Once tested, the change is deployed according to the **Deployment Plan**.
    - All project documents (SRS, SPMP, user guides) are updated to reflect the change.
6. **Monitoring**
    - The team checks that the change works as intended in the live system.

Example:

- If a new AI logic is added to improve itinerary suggestions, the team submits a change request, evaluates the effect on schedule, approves it, implements and tests it, and then updates the documentation.
- If the **Itinerary Sharing** feature needs to include a QR code fix, it follows the same process.

This process ensures that all modifications are controlled, tested, and documented while keeping the project on track and organized.

4 SCHEDULE/TIME MANAGEMENT

The **PhiliFinds project** schedule will follow a structured timeline to ensure timely completion of all modules, including **Dynamic Itinerary Builder, Cost Estimator, Itinerary Sharing, Navigation Assistance, Notifications, and Emergency Support**. The Project Manager (PM) will be responsible for maintaining the schedule, monitoring progress weekly, and updating it whenever there are changes in scope or unforeseen delays.

Schedule Management Approach

- A baseline schedule will be established at the start of the project.
- Progress will be monitored weekly, and variances greater than 10% on key milestones will be addressed immediately.
- All tasks will include dependencies to avoid blocking critical features, such as AI-based itinerary generation before cost estimation.
- Any changes will be handled using the **Change Control Management** process outlined in Section 3.3.

4.1 MILESTONES

The table below lists the milestones for this project, along with their estimated completion timeframe.

Milestones	Estimated Completion Timeframe
Project Approval & Requirement Analysis	Week 1–2
UI/UX Design Completion (Wireframes & PWA Layout)	Week 3–4
Backend Setup (Supabase DB, Authentication, RLS)	Week 4–5

Milestones	Estimated Completion Timeframe
AI Engine Development for Itinerary Generation	Week 5–7
Frontend Development (Itinerary Builder, Cost Estimator, Sharing Module, Notifications)	Week 6–8
Integration and Module Testing (All Features including Itinerary Sharing)	Week 8–9
Deployment to Staging and User Acceptance Testing	Week 9–10
Final Production Deployment	Week 10–11
Documentation & Project Submission	Week 11–12

4.2 PROJECT SCHEDULE

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4.3 Dependencies

- **Design-to-Frontend**: Frontend Development cannot reach its final stages without the completed UI/UX mockups
- **API Availability:** Project progress depends on the uptime and response speed of third-party AI and Mapping APIs
- **AI-to-Frontend**: The Dynamic Itinerary Builder UI depends on the API integration for AI itinerary generation to display real data

5 COST/BUDGET MANAGEMENT

The PhiliFinds development team will implement a structured cost management approach to ensure financial sustainability, efficient resource allocation, and controlled operational expenses throughout the project lifecycle.

As a startup development group composed of Computer Engineering students, the team will apply industry-standard budgeting practices, including cost estimation, budget tracking, variance analysis, and contingency planning.

Cost Management Approach

- A baseline budget will be established during project initiation.
- All projected expenses will be documented and approved by the Project Manager and Finance Lead.
- Monthly cost monitoring will be conducted to track variances.
- Any cost variance exceeding 10% of the allocated budget will require management review.
- A contingency reserve will be allocated for unexpected technical or operational expenses.

Category	**Description**	**Duration**	**Cost (₱)**
Personnel	Project Manager	3 Months	105,000
Personnel	Backend Developer	3 Months	90,000
Personnel	Frontend Developer	3 Months	90,000
Personnel	UI/UX Designer	2 Months	50,000
Infrastructure	Cloud Services & Hosting	3 Months	11,100
Infrastructure	Domain & Software Tools	Annual / 3 Months	3,000
Operations	Marketing & Utilities	Project Duration	16,000
Operations	Contingency Reserve	Project Duration	32,000
TOTAL PROJECT BUDGET			**₱397,100**

6 QUALITY MANAGEMENT

The PhiliFinds development team will implement formal Quality Assurance (QA) and Quality Control (QC) processes to ensure that all system requirements are met with high reliability, performance, and usability standards. Quality will be maintained throughout the Software Development Life Cycle (SDLC) through reviews, testing, and structured defect management.

Quality Objective	Target Standard	Validation Method
Functional Correctness	All critical features operate as specified	Functional & System Testing
Defect Prevention	No critical or high-severity defects before release	Defect Tracking & Retesting
UI Consistency	Uniform screen layouts and navigation flow	UI Inspection & Usability Review
Accurate Calculations	Cost estimates and itinerary data computed correctly	Unit Testing & Integration Testing
Performance Efficiency	Acceptable load time and responsive interface	Performance Testing
Requirement Compliance	100% of critical requirements implemented	Requirements Traceability Matrix (RTM)

7 HUMAN RESOURCE MANAGEMENT

The PhiliFinds development team will implement a structured human resource management plan to ensure clear role definition, accountability, and efficient collaboration throughout the project lifecycle. Team members will be assigned responsibilities based on expertise in

software development, system design, and project management.

Role	Key Responsibilities
Project Manager	Oversees project execution, schedule, budget, and stakeholder communication
Backend Developer	Develops database, APIs, authentication, and server-side logic
Frontend Developer	Implements user interface, PWA functionality, and system integration
UI/UX Designer	Designs wireframes, user flows, and ensures interface consistency
QA / Test Engineer	Conducts testing, manages defect tracking, and validates requirements

7.1 STAFFING PLAN

- Core team members will be engaged throughout the 3-month development phase.
- The UI/UX Designer will primarily support early design phases and assist during usability testing.
- QA activities will intensify during integration and pre-deployment stages.

7.2 ROLES AND RESPONSIBILITIES MANAGEMENT

- Each team member is accountable for assigned deliverables.
- Weekly progress meetings will be conducted to monitor performance.
- Performance will be evaluated based on milestone completion and quality standards.
- Any resource gaps will be addressed through workload reallocation or temporary support engagement.

8 COMMUNICATIONS MANAGEMENT

The PhiliFinds project team will implement a structured communication plan to ensure timely information sharing, coordination, and transparency among stakeholders. Communication will support project monitoring, issue resolution, and decision-making throughout the development lifecycle.

Regular meetings, progress reports, and collaborative tools will be used to maintain alignment between the project manager, development team, and stakeholders. All critical updates, risks, and milestone progress will be communicated through agreed communication channels.

Communication Type	Purpose	Medium	Frequency
Project Meetings	Discuss progress, tasks, and issues	Video Conference / Meeting	Weekly
Progress Reports	Update stakeholders on project status	Written Report / Email	Bi-weekly
Team Coordination	Day-to-day development communication	Team Messaging Platform	Daily
Milestone Review	Evaluate completion of major deliverables	Presentation / Meeting	Per Milestone
Issue Escalation	Address critical risks or blockers	Direct Communication	As Needed

Communication Guidelines

- The Project Manager is responsible for coordinating all formal communications.
- All major project updates must be documented and shared with stakeholders.
- Communication tools will be used to ensure transparency and efficient collaboration.
- Critical risks or issues will be escalated immediately to the project manager for resolution.

8.1 COMMUNICATION MATRIX

Stakeholder	Messages	Vehicles	Frequency	Communicators	Feedback Mechanisms

Project Sponsor / Investors	Project progress, budget status, milestone completion, risks	Formal Report, Presentation, Email	Monthly / Per Milestone	Project Manager	Review meetings, approval confirmations
Project Manager	Task updates, risk reports, technical issues, schedule status	Team Meetings, Messaging Platform	Weekly / As Needed	Development Team	Status discussions, issue tracking system
Development Team (Backend, Frontend, UI/UX)	Task assignments, integration updates, technical clarifications	Messaging Platform, Code Repository, Meetings	Daily	Project Manager / Team Leads	Stand-up meetings, pull request reviews
QA / Test Engineer	Test results, defect reports, validation status	Defect Tracking System, Email	Per Testing Cycle	QA Engineer	Retesting results, defect closure confirmation
Clients / End Users	Feature updates, release announcements, support information	Email, Release Notes, System Notifications	Per Release	Project Manager / Support Team	User feedback forms, surveys, issue reports

9 RISK MANAGEMENT

The PhiliFinds development team will implement a structured risk management process to proactively identify, assess, monitor, and mitigate risks that may affect project scope, schedule, cost, quality, or performance. Risk management activities will be conducted throughout the entire project lifecycle.

Risk Management Approach

The project will follow a continuous risk management cycle consisting of:

1. **Risk Identification**
   Potential technical, operational, financial, and external risks will be identified during project planning and ongoing development activities.

2. **Risk Analysis and Assessment**

Each identified risk will be evaluated based on its probability of occurrence and potential impact on the project. Risks will be categorized as Low, Medium, or High priority.

3. **Risk Response Planning**
   Appropriate mitigation, avoidance, transfer, or acceptance strategies will be developed for significant risks. Preventive measures will be prioritized over reactive solutions.

4. **Risk Monitoring and Control**
   Risks will be reviewed during weekly project meetings. New risks will be added as they arise, and mitigation strategies will be adjusted as needed.

Roles and Responsibilities

- The Project Manager is responsible for overseeing the risk management process.
- Team members are responsible for reporting newly identified risks.
- High-priority risks will be escalated immediately to project leadership for decision-making.

9.1 RISK LOG

Risk_Logs

10 ISSUE MANAGEMENT

The PhiliFinds development team will implement a structured issue management process to ensure that project-related problems are identified, documented, tracked, resolved, and formally closed in a timely manner. Issue management focuses on active problems that require immediate attention, unlike risks which are potential future events.

Issue Management Approach

The project will follow a controlled issue resolution process consisting of:

1. **Issue Identification**
   Any team member may identify and report an issue affecting project scope, schedule, cost, quality, or performance.
2. **Issue Documentation**
   All issues will be recorded in an Issue Log, including description, date identified, severity level, responsible owner, and target resolution date.
3. **Issue Assessment and Prioritization**
   Issues will be classified as Low, Medium, or High severity based on impact and urgency.
4. **Issue Resolution**

Assigned team members will implement corrective actions. High-severity issues will be escalated to the Project Manager immediately.

5. **Issue Closure**
   An issue will be formally closed once resolution has been verified and documented.

Roles and Responsibilities

- The Project Manager oversees issue tracking and ensures timely resolution.
- Assigned team members are responsible for implementing corrective actions.
- All team members are responsible for reporting issues promptly.

10.1 ISSUE LOG

Issue_Logs

11 PROCUREMENT MANAGEMENT

[Example: This can include information such as ensuring project team members are assigned computers, how development and test servers are procured or can go into more detail and include an acquisition strategy that details how the project will be staffed (e.g., performance based fixed price contract, CITS contractors).]

The PhiliFinds development team will implement a structured procurement management process to acquire the necessary hardware, software, cloud services, and external resources required for successful project execution. Procurement activities will ensure that all required resources are obtained in a timely and cost-effective manner while maintaining quality standards.

Procurement Approach

- Procurement requirements will be identified during project planning.
- The Project Manager will review and approve all procurement requests.
- Preference will be given to cost-efficient and scalable solutions.
- Contracts or subscriptions will be evaluated based on cost, reliability, and long-term sustainability.

Procurement Categories

Category	Description	Procurement Strategy

Hardware	Developer workstations, testing devices	Company-issued equipment or approved purchase
Cloud Services	Database, authentication, hosting services	Subscription-based cloud providers
Software Tools	Development tools, design software, collaboration platforms	Licensed or subscription-based plans
Domain & Deployment	Domain registration, hosting configuration	Annual domain registration and managed hosting
External Services (if required)	Third-party APIs, consultants, or contractors	Fixed-price or short-term service agreement

Procurement Control

- All purchased services will be documented and monitored for usage and cost.
- Subscription renewals will be reviewed before extension.
- Vendor performance will be evaluated based on service reliability and support responsiveness.
- Procurement records will be maintained for audit and financial tracking purposes.

12 COMPLIANCE RELATED PLANNING

Compliance Area	Requirement	Action Plan
Data Privacy	Philippine Data Privacy Act (DPA) of 2012	Implement Row-Level Security (RLS) in the database.
Cybersecurity	SSL/TLS Encryption	All production traffic on Vercel must use HTTPS; API keys must be kept server-side.
Accessibility	W3C WCAG 2.1 Standards	Ensure UI/UX designs follow accessibility guidelines for diverse users.
User Safety	Emergency Information Accuracy	Regularly verify that displayed emergency hotlines are current and functional.

Appendix A: Project Management Plan Approval

The undersigned acknowledge they have reviewed the **PhiliFinds Project Management Plan** and agree with the approach it presents. Changes to this **Project Management Plan** will be coordinated with and approved by the undersigned or their designated representatives.

Signature: _____ Date: _____
Print Name: Jacob Isaac B. Enriquez
Title: Project Manager
Role:

Signature: _____ Date: _____
Print Name: Alexa F.Atayde
Title: Student Developer
Role:

Signature: _____ Date: _____
Print Name: Nadine Kate P. Ygot
Title: Student Developer
Role:

Signature: _____ Date: _____
Print Name: Marley Obert P. Mariquit
Title: Student Developer
Role:

Signature: _____ Date: _____
Print Name: Kerstan Zam M. Davide
Title: Student Developer
Role:

APPENDIX B: REFERENCES

The following table summarizes the documents referenced in this document.

Document Name and Version	Description	Location

APPENDIX C: KEY TERMS

The following table provides definitions for terms relevant to this document.

Term	Definition